Your first analysis

NSDF-Data student guide 2 — macOS edition

Version 22 · 19 October 2026

Tip

What you will do. Download one set of real detector data, open it in a notebook, look at it, and reproduce one number from an existing note. You will need the setup guide for macOS finished first.

Before you start

- **Time:** about two hours.
- **You need:** the setup guide finished (your environment `darkmatter_cli_env` exists and works).
- **Check:** open a terminal (Step 1 of the setup guide), and run `conda activate darkmatter_cli_env`. The start of the line should show `(darkmatter_cli_env)`.
- **Folder:** you do not need to be in any particular folder yet. Each step below tells you where to go.

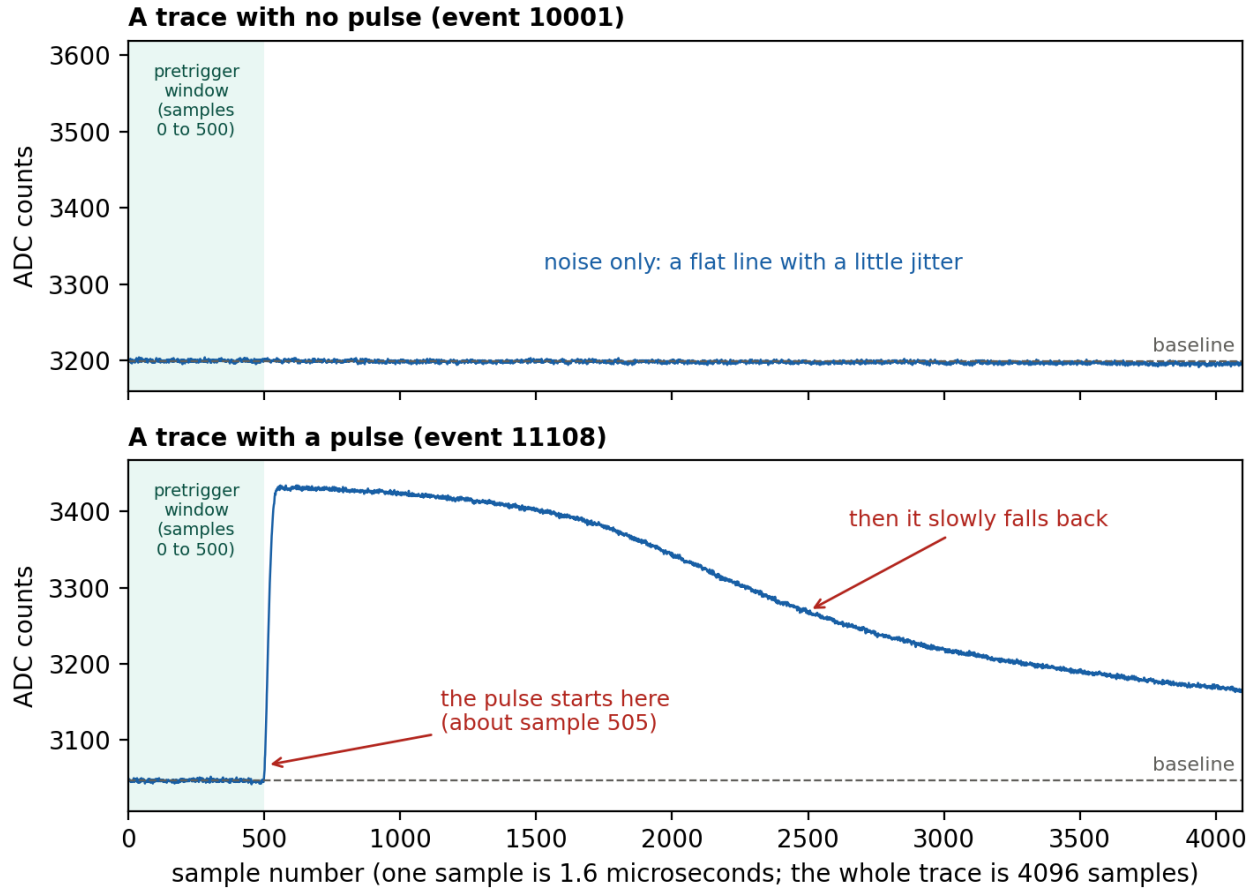
Routine maintenance

Keep your setup up to date. The first code cell in Step 4 checks your environment against the project's current list and warns you if it is out of date. If you see a WARNING, follow Updating your setup. Do the same after every new release the project lead announces.

Words you will meet

Word	What it means here
Series	One data-taking session, named like `07221203_2025`. The first two digits are the test facility (07 is UMN, the University of Minnesota); the rest are the date and time the series started, here 221203 (year, month, day: 3 December 2022) and 2025 (20:25). A typical series lasts hours; some are shorter because they are tests.
Dump	One file set taken from a series, numbered `F0001`, `F0002`, …

Word	What it means here
Event	One moment when the system decided something might have happened and saved a short recording from every detector. Events are numbered in time order. See **Figure 2**, and the appendix.
Detector	In this data, a “detector” number (0, 1 or 2) means one readout electronics unit, not a separate crystal. All three units are wired to the same physical detector, S104.
Channel	One of the four sensor outputs of a readout unit, numbered 0 to 3. Each channel records its own trace. We use detector 0, channel 0 (**Figure 2**).
Trace	The recorded signal for one channel in one event: 4096 numbers, one per sample.
Sample	One reading of the readout circuit’s output (usually a voltage or a current). Each sample is 1.6 microseconds long, and is the average of the output over that time.
ADC counts	The units of a sample. The analog-to-digital converter (ADC) turns each reading into a whole number (an integer), proportional to the measured voltage or current. In this dump the largest are around 8000, so they need at least 13 bits; the exact width is not stated in this guide.
Baseline	The flat level a trace sits at when nothing happens.
Pretrigger window	The first samples of a trace, used to measure the baseline.
Pulse	A sudden rise above the baseline that then slowly falls back. It is what a particle leaves in a trace.



Series 07221203_2025, dump F0001, Detector 0, Channel 0. Both panels show the same height in counts.

Figure 1. Two traces on the same scale. Top: noise only. Bottom: a pulse.

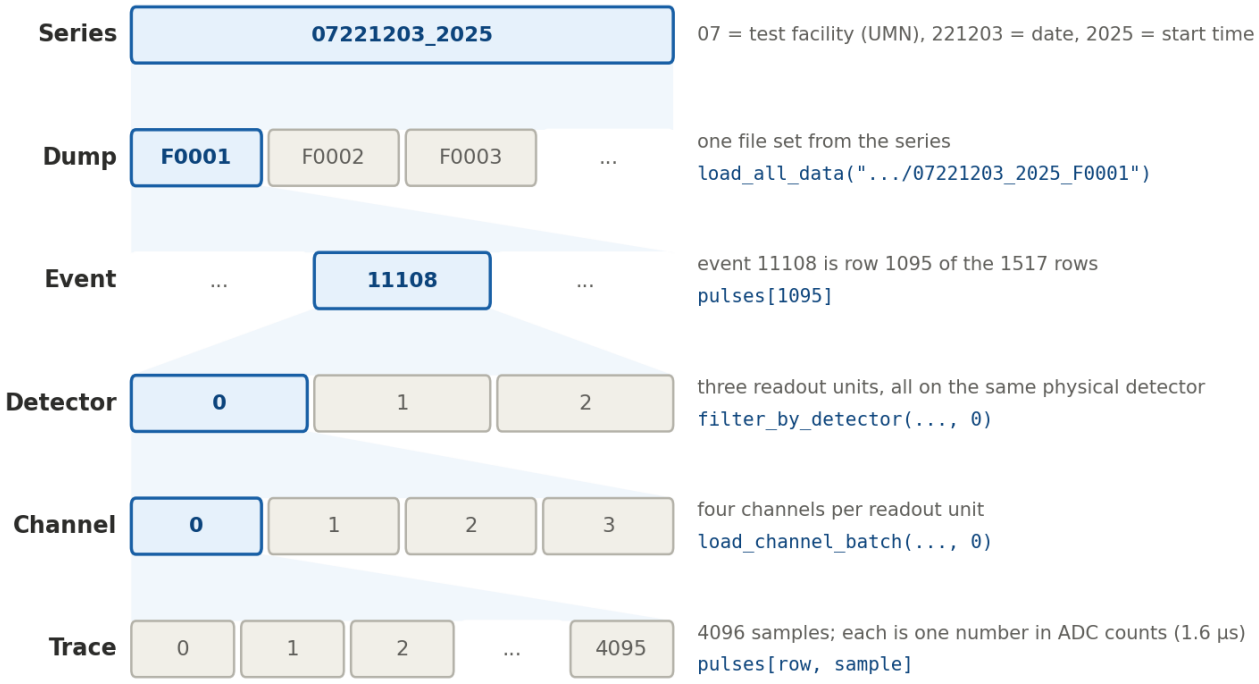


Figure 2. How the data is organised, and which part this guide follows (blue). A series is split into dumps; a dump holds events; each event has a trace from every channel of every detector; a trace is 4096 samples. The code in Step 4 picks detector 0, channel 0 from every event of dump `F0001`, so each row of the table `pulses` is one event, and `pulses[row, sample]` is one number.

Step 1. Read two notes first (about 30 minutes)

Open the notes site at villano-lab.github.io/NSDF-Data and read these, in order:

1. **Note 2**, *Noise-pulse selection*: how we pick quiet traces. It uses the same dump you will download.
2. **Note 2a**, *bstd against time*: how the baseline noise of those traces changes from event to event, in the same dump.

You do **not** need to follow every number. Just learn the words in the table above.

Step 2. Download the data

Navigation

Where am I, and what is here? Two commands show you. Try both now; neither changes anything.

- `pwd` prints the folder you are in. The prompt also names it.
- `ls` lists the files and folders inside it.

To move, type `cd` and a folder name from the list, or `cd ..` to go back up one level.

First go to your home folder, so the data lands in the right place. Type this, then press Enter:

```
cd ~
```

Now download the data:

```
nsdf-cli download 07221203_2025_F0001
```

The download puts the data in an `idx` folder **in the folder you are in**. Because you went to your home folder first, it lands in `~/idx`, inside a folder named `07221203_2025_F0001`. The data is about 34 MB. Wait until the prompt comes back.

Careful

Never download from inside the `NSDF-Data` folder. Its `idx` folder would sit inside the project, where Git can pick it up. If you have already downloaded there, move the `idx` folder to your home folder, and tell your lead.

What these lines do: the first one moves you to your home folder (the `cd` line). The second, `nsdf-cli download`, fetches the data set from the NSDF servers over the internet and puts it in a new `idx` folder.

Tip

Optional: tell `nsdf-cli` once where downloads go. Type this one time, and every later download goes to your `idx` folder, whichever folder you are in:

```
nsdf-cli config --output-global ~/idx
```

Type `nsdf-cli config` to see the setting. It is saved in a small file in your home folder (`.nexus-dm/config.json`), so you do not repeat it. Two more `nsdf-cli` options you may see: `nsdf-cli download SERIES --all` downloads **every** dump of a series (the series in this guide has 30 dumps, about 1.9 GB), so do not use `--all` unless your lead says to; and `nsdf-cli prune` removes the downloaded series from your download folder to free disk space (it shows the folder and asks before it deletes anything).

Careful

Do not move or edit the downloaded files, and do not add them to Git. They are external data, not part of our data analysis code in the NSDF-Data repository.

Tip

If the message says the folder already exists, the data may already be there. To check, type `ls ~/idx`: if the list shows a folder named `07221203_2025_F0001`, you already have it, and an error message means there is no `idx` folder yet. Ask the project lead before you delete anything.

Step 3. Open Jupyter in the right folder

Every student works in a folder of their own, inside `data_analysis/R76/student` in the project. Use the same name as in your branch name (`student-yourname`), without spaces, in place of `yourname`. In the terminal, type these lines:

```
conda activate darkmatter_cli_env
cd ~/Research/NSDF-Data/data_analysis/R76/student
mkdir yourname
cd yourname
jupyter lab
```

What these lines do: `conda activate` switches your environment on, `cd` moves you into the folder where the students' notebooks live, `mkdir yourname` makes your own folder there (if it says the folder already exists, that is fine: you made it before), `cd yourname` moves you into it, and `jupyter lab` starts the notebook program. **Leave this terminal window open while you work**: closing it stops Jupyter.

Your web browser opens a Jupyter page. Click **File**, then **New**, then **Notebook**. If it asks which kernel to use, choose **darkmatter_cli_env**.

Tip

A notebook is a list of boxes called **cells**. Type code in a cell and press **Shift + Enter** to run it. The result appears under the cell. Press **Enter** (without Shift) to keep typing in the same cell.

Give your notebook a name: right-click its tab at the top (it says `Untitled.ipynb`) and choose **Rename Notebook**. In the box, replace `Untitled` with `first-analysis-yourname` (keep `.ipynb` at the end) and click **Rename**.

Step 4. Load the data and count the traces

Copy this into the first cell and press **Shift + Enter**:

```
import sys
from pathlib import Path

# Tell Python where the pulse library (the python folder) is.
# This looks upward from the notebook's folder until it finds the project,
# so it works wherever in the project your notebook is.
here = Path.cwd().resolve()
REPO = next((p for p in [here, *here.parents] if (p / "python" / "pulse_io.py").exists
assert REPO, "Start Jupyter from inside the NSDF-Data folder."
sys.path.insert(0, str(REPO / "python"))

from nsdf_dark_matter.idx import load_all_data
import pulse_io as pio

# Check that your setup is up to date (it only warns; it never stops the notebook).
import check_environment
check_environment.check()

# Open the dump you downloaded.
cdms = load_all_data(str(Path.home() / "idx" / "07221203_2025_F0001"))

# Take every trace from detector 0, channel 0, into one table.
ids, pulses = pio.load_channel_batch(
    cdms, pio.filter_by_detector(cdms.get_detector_ids(), 0), 0)

print(pulses.shape)
```

Checkpoint

The first line says `Environment OK`, and the last is `(1517, 4096)`: 1517 traces, each with 4096 samples. If you get a different number, stop and check the dump name and the folder in Step 2. If the first line says `WARNING`, you can keep going for now, but follow *Updating your setup* soon.

Want to know what every line did? See the code, line by line (part A).

Step 5. Look at a trace

Click in the empty cell under the one you just ran: Jupyter adds one each time you press **Shift + Enter**. If there is none, click **+** in the toolbar to add one. Put this in it and run it:

```
import matplotlib.pyplot as plt

plt.plot(pulses[0])
plt.xlabel("sample (1.6 microseconds each)")
plt.ylabel("ADC counts")
plt.show()
```

Now change the 0 to 1 and run again, then try 2 and 3, one number at a time. (`pulses[1,2,3]` does not pick three traces: it is an error, because the first number counts rows and the others count samples.) Most traces look like flat noise, with a little jitter. Row 1 is a quiet trace like the top one in **Figure 1**. To see a pulse, try row 1095. Then see the appendix if you want to know more about what you are looking at.

Step 6. Reproduce a number from Note 2

Note 2 says that for this dump, a strict selection leaves **179 quiet traces**. Run this in the next empty cell (add one with **+** if there is none):

```
from dataclasses import replace
from pulse_config import DEFAULT_CONFIG
import pulse_quantities as pq
import pulse_cuts as pc
import numpy as np

c500 = replace(DEFAULT_CONFIG, pretrigger_samples=500)
quiet = pc.excursion_band(pulses, c500) & (np.log10(pq.bstd(pulses)) < 0.5)
print(quiet.sum())
```

Checkpoint

The output is **179**. If you see a different number, that is worth knowing. Write down the number you got and the steps you took, and tell the project lead. Do not change the project's code to make the number match.

Want to know what every line did? See the code, line by line (part B).

Step 7. Write down what you did

In Jupyter, click **+** to add a cell and change its type from *Code* to **Markdown** in the toolbar. In that cell, write a few sentences in plain English: what you loaded, what you plotted, and what number you got. Then press **Shift + Enter**.

Save with **Command + S** (macOS) or **Ctrl + S** (Windows and Linux).

Once your branch exists (setup guide, Step 8), save your notebook to Git. The first terminal is busy: it is still running Jupyter, so leave it open. Open a **new** terminal for the Git commands. A new terminal starts in your home folder, so the first line below moves you into the project folder from the setup guide:

```
cd ~/Research/NSDF-Data
git branch --show-current
git status
git add data_analysis/R76/student/yourname/first-analysis-yourname.ipynb
git commit -m "First analysis: load dump 1 and count the quiet traces"
git push -u origin student-yourname
```

What these lines do: the first one moves you into the project. `git branch --show-current` prints the name of the branch you are on, and `git status` lists what has changed since your last

save. **Make this a habit: every time you open a terminal in a project, look at the branch and the status before you do anything else.** Then `git add` picks the file you want to save, `git commit` saves a snapshot of it with your short message, and `git push` uploads that snapshot to GitHub, to your own branch.

Checkpoint

`git branch --show-current` prints `student-yourname` (your own branch), and `git status` starts with `On branch student-yourname` and lists your notebook under **Untracked files** (a new file) or **Changes not staged for commit** (a file you changed). If the branch is `develop` or `master`, stop and ask the project lead: do not commit there.

Careful

Commit only your notebook. Check the list from `git status` before committing. If you see `.bin` files or anything from the `idx` folder, do not add them.

You are done when

- ☐ `(1517, 4096)` printed after loading the data.
- ☐ You plotted at least three traces and can say what the flat ones look like.
- ☐ You got 179 from the selection, or wrote down what you got and why you think it differs.
- ☐ Your notebook is saved, and committed to your branch if you have access.

Common problems

What you see	What to do
`ModuleNotFoundError: No module named 'nsdf_dark_matter'`	You are not in the environment. Run `conda activate darkmatter_cli_env`, then restart the notebook kernel (**Kernel**, then **Restart Kernel**).
`ModuleNotFoundError: No module named 'pulse_io'`	Jupyter was started from the wrong folder. Close it and start it again from your own folder in `data_analysis/R76/student` (Step 3).
`fatal: pathspec '...' did not match any files` when you run `git add`	You are not in the project folder, or the file name is not the one you saved. Open a terminal (Step 1 of the setup guide), run `cd ~/Research/NSDF-Data`, then `git status`, and copy the notebook's path exactly as Git prints it into `git add`.
`IndexError: too many indices for array`	You wrote something like `pulses[1,2,3]`. Put one number in the brackets, for example `pulses[1]`, and run the cell again.
`FileNotFoundError` on the dump folder	The download did not finish, or it went somewhere else. Check the folder in Step 2.

What you see	What to do
A huge spike at the very start of traces	This is a real electronics glitch in the first few samples, not a bug. It is explained in Note 2a.
The kernel is not `darkmatter_cli_env`	Click the kernel name at the top right and choose **darkmatter_cli_env**.

Careful

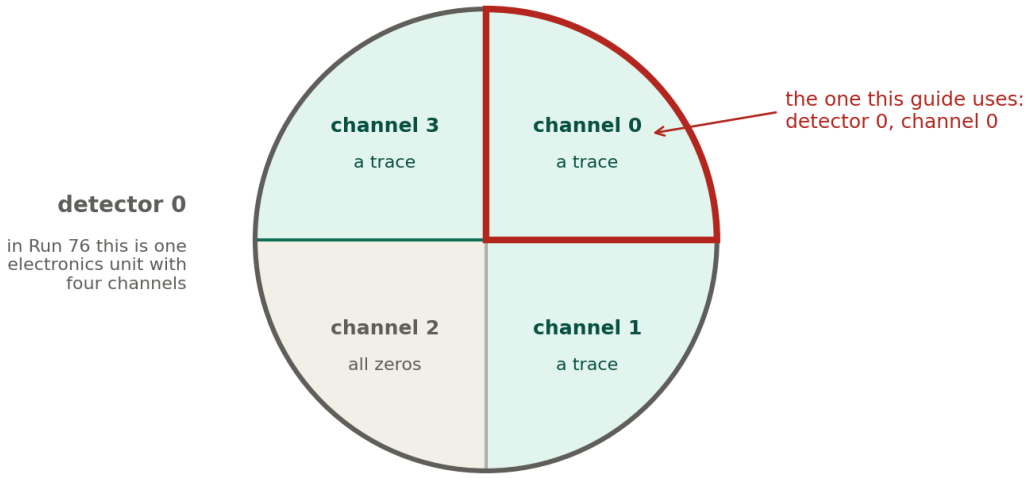
Do not run `07221203_2025_dump1_noise.ipynb` unless the project lead says it is fine. It overwrites a shared archive file.

Appendix: what an event looks like

You do not need this to finish the guide. It explains the words in the table at the top.

The detector system watches its sensors all the time. Whenever it decides that something may have happened, it saves a short recording from every sensor. That moment is one **event**. Events are numbered in time order, and the numbers restart in every dump, so an event number alone does not identify an event.

In Run 76 one physical detector (a silicon detector, S104) is read out by three older ("legacy") electronics units, each with four **channels**, all connected to that same detector. The software calls the units detector 0, 1 and 2, and so does this guide. One event therefore holds up to 3 x 4 = 12 recordings. Each recording is a **trace**: 4096 numbers, one per sample. In this dump, 7 of the 12 slots hold a trace and the other 5 read all zeros. We follow one channel, detector 0 channel 0, through all 1517 events.



One event. The data has 3 units of 4 channels (12 slots); in Run 76 all 3 read the same physical detector. 7 slots hold a trace of 4096 samples, the rest read zero. A schematic, not a real channel map.

Figure 3. The four channels of detector 0 in one event. It is a schematic: the four sectors only stand for the channel numbers, and it does not show where the real sensors sit. The detector's own sensor layout is similar to the SuperCDMS HV detector mask, but not the same.

A trace looks like a flat line with a little jitter (the **baseline**) until something happens. The first 500

samples, the **pretrigger window**, are used to measure the baseline. If a particle deposits energy, the trace jumps up at once and then slowly falls back: a **pulse**. In **Figure 1**, the pulse in event 11108 starts just after sample 500; in this dump, many pulses start there.

The pulse trace in **Figure 1** is row 1095 of the table `pulses` from Step 4 (the row is the position in the table, not the event number). Row 1 is a quiet trace.

Appendix: the code, line by line

You do not need this to finish the guide. It goes through the code you pasted, one line at a time: each line of code, and next to it, in red, what it does. A line that begins with # is a comment: Python ignores it, and it is there for people to read, so the comments are left out here.

Part A. Step 4: load the data and count the traces

```
import sys
from pathlib import Path
```

↑ Bring in two tools. `sys` controls where Python looks for code. `Path` works with folder names in the same way on every kind of computer.

```
here = Path.cwd().resolve()
```

↑ The folder Jupyter is working in, written out in full.

```
REPO = next((p for p in [here, *here.parents]
             if (p / "python" / "pulse_io.py").exists()), None)
```

↑ Look at `here` and then at each folder above it, one after another, and take the first one that contains the file `python/pulse_io.py`. That folder is the whole NSDF-Data project. If there is none, `REPO` is `None`.

```
assert REPO, "Start Jupyter from inside the NSDF-Data folder."
```

↑ Stop with that message if the project was not found, which means Jupyter was started in the wrong place.

```
sys.path.insert(0, str(REPO / "python"))
```

↑ Tell Python to look in the project's `python` folder first, so that the next `import` lines can find the project's own code.

```
from nsdf_dark_matter.idx import load_all_data
```

↑ Bring in the function, from the NSDF software, that reads a downloaded dump.

```
import pulse_io as pio
```

↑ Bring in the project's helpers for traces. `as pio` gives it a short name.

```
import check_environment
check_environment.check()
```

↑ Bring in the project's environment check and run it. It compares your Python and your packages with the project's list (`requirements.txt`) and prints `Environment OK`, or a `WARNING` that says what differs and where to read how to fix it. It is only a warning: your notebook carries on either way.

```
cdms = load_all_data(str(Path.home() / "idx" / "07221203_2025_F0001"))
```

↑ Open the dump you downloaded in Step 2. `Path.home()` is your home folder, and `/ "idx" / "07221203_2025_F0001"` follows the path down to the dump, the same folder Step 2 created. The whole dump is read into the computer's memory and called `cdms`.

```
ids, pulses = pio.load_channel_batch(
    cdms, pio.filter_by_detector(cdms.get_detector_ids(), 0), 0)
```

↑ Read it from the inside out. `cdms.get_detector_ids()` is the list of names of every group of traces (a name holds an event number and a detector number). `pio.filter_by_detector(..., 0)` keeps only the names for detector 0. `pio.load_channel_batch(cdms, ..., 0)` takes channel 0 from each of those and stacks the traces into one table. It gives back two things: `ids`, the names in order, and `pulses`, the table, with one row for each event and one column for each sample (see **Figure 2**).

```
print(pulses.shape)
```

↑ Show the size of the table as (rows, columns). That is the `(1517, 4096)` you checked.

Part B. Step 6: count the quiet traces

```
from dataclasses import replace
```

↑ A tool for making a copy of a group of settings with one value changed.

```
from pulse_config import DEFAULT_CONFIG
```

↑ The project's standard settings, such as how many samples at the start of a trace count as the baseline window (1000 by default) and how many of the very first samples to skip because of the electronics glitch (10).

```
import pulse_quantities as pq
```

↑ Functions that work out one number for each trace, such as how much its baseline wobbles.

```
import pulse_cuts as pc
```

↑ Functions that answer yes or no for each trace: "keep this one?"

```
import numpy as np
```

↑ Tools for working with tables of numbers.

```
c500 = replace(DEFAULT_CONFIG, pretrigger_samples=500)
```

↑ A copy of the standard settings in which the baseline window is the first 500 samples, as in Note 2.

```
quiet = pc.excursion_band(pulses, c500) & (
    np.log10(pq.bstd(pulses)) < 0.5)
```

↑ Two yes-or-no tests joined by `&`, which means "and". The result, `quiet`, has one `True` or `False` for each trace. `pc.excursion_band(pulses, c500)` is `True` when the trace's largest swing after the baseline window, compared with the wobble of its own baseline, falls in the middle band chosen in Note 2. `pq.bstd(pulses)` is the wobble (the standard deviation) of each trace's baseline, and `np.log10(...) < 0.5` keeps only the traces whose baseline wobbles little: under about 3 counts.

```
print(quiet.sum())
```

↑ A `True` counts as 1 and a `False` as 0, so the sum is the number of traces that passed both tests. That is the 179 of the checkpoint.

Where to get help

Anthony Villano, project lead: anthony.villano@ucdenver.edu

When you write, include the step number, the command you typed, and the last few lines the computer printed. A screenshot helps. Nobody will be annoyed by a question about a step that did not work.

Session Info

Guide version 22, 19 October 2026, macOS edition. Written for beginners; please tell the project lead where a step was unclear.

Key links

- Note 2: https://villano-lab.github.io/NSDF-Data/notes/note-02-noise-selection.html
- Note 2a: https://villano-lab.github.io/NSDF-Data/notes/note-02a-bstd-vs-time.html
- Note 1 (series catalog) and Note 1a (python library structure), for later: https://villano-lab.github.io/NSDF-Data/
- Repository: https://github.com/villano-lab/NSDF-Data